

Reproducing SIDReasoner: Reasoning over Semantic IDs

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Abstract. SIDReasoner represents catalogue items as Semantic IDs (SIDs) and trains an LLM to recommend by generating SID sequences. We reproduce it on Amazon Video Games, Office Products, and Industrial & Scientific, focusing on whether GRPO reinforcement learning improves the reasoning-activated checkpoint over the base stage. We also extend the pipeline on Office Products by varying reward design, roll-out constraints, LoRA updates, and inference-time reasoning, and test transfer to a sampled Yelp Restaurants domain. Our reproduction recovers the main trend: RL fine-tuning improves over the prior stage on all Amazon datasets, though scores trail the paper on most HR metrics; the strongest gain is on Video Games, where ranking quality exceeds the paper’s reported NDCG. Our extensions show unconstrained RL exploration followed by constrained inference outperforms constrained rollouts throughout training, while inference-time reasoning lowers ranking quality and increases runtime. On Yelp, RL fine-tuning does not improve over the base stage, and both checkpoints underperform SASRec. Overall, grounded SIDs and outcome-based RL are useful, but reasoning and transfer remain sensitive to training, decoding, and dataset choices.

GitHub Link: <https://github.com/erinelgz/SIDReasoner-UvA>

Keywords: Generative recommendation · Semantic IDs · Reproducibility · Reinforcement learning

1 Introduction

Sequential recommendation predicts a user’s next interaction from previous behaviour. Recent generative recommenders reformulate this task as autoregressive sequence generation, allowing language-model architectures to predict items directly. A central design choice is item representation. Textual item descriptions preserve semantic information but are expensive to decode and difficult to map back to catalogue items. Sparse item identifiers are compact and grounded but semantically opaque. Semantic IDs (SIDs) address this trade-off by representing each item as a short sequence of discrete tokens derived from semantic item embeddings [7, 12].

SIDReasoner [5] extends SID-based generative recommendation by training a language model to reason over item-level SID tokens. This is challenging because newly introduced SID tokens lack inherent meaning, while recommendation reasoning is difficult to supervise. The original method therefore strengthens

SID–language alignment through multi-task fine-tuning and teacher-generated semantic enrichment, followed by outcome-driven reinforcement learning with Group Relative Policy Optimization (GRPO) [13]. The released code consists of three training stages: SID–language alignment, reasoning activation, and GRPO-based reinforcement learning. We refer to the reasoning-activated checkpoint as Stage 2 and the RL-updated checkpoint as Stage 3.

We study SIDReasoner because it combines custom SID vocabularies, enriched alignment data, reasoning activation, reinforcement learning, constrained decoding, and full-item ranking evaluation. Rather than only comparing final scores, we identify which parts of the method remain robust under realistic academic reproduction. We address four research questions: **RQ1**, whether the reported SIDReasoner results and Stage 3 improvement can be reproduced on the original Amazon datasets; **RQ2**, how reward design, rollout constraints, and LoRA updates affect Office Products ranking quality; **RQ3**, whether explicit inference-time reasoning improves recommendation or can be used selectively at lower cost; and **RQ4**, whether the SIDReasoner pipeline transfers to a sampled Yelp Restaurants domain.

Our reproduction recovers the paper’s main qualitative trend: Stage 3 improves over Stage 2 on all three Amazon datasets. However, the reported scores are not fully matched. The gap is largest on Office Products and smaller on Industrial & Scientific, while Video Games shows lower HR but higher NDCG than reported. Our extensions show that unconstrained RL exploration followed by constrained inference improves Office Products performance, whereas generated reasoning at inference consistently reduces ranking quality and increases runtime. Finally, transfer to Yelp Restaurants is unsuccessful: Stage 3 does not improve over Stage 2, and SASRec remains stronger. These findings support SID grounding and outcome-based RL, but show that the benefits of explicit reasoning and domain transfer are not automatic.

2 Related Work

Generative recommendation and Semantic IDs. Sequential recommendation has traditionally been studied with discriminative models such as GRU4Rec, Caser, and SASRec [6, 16, 11]. Generative recommendation instead predicts the next item autoregressively [9]. Sparse-ID methods are compact but semantically weak, while text-based methods exploit item semantics but suffer from decoding and grounding difficulties [4, 1, 2]. SID-based methods address this trade-off by representing each item as a short sequence of semantic tokens. TIGER introduced generative retrieval with vector-quantized item IDs [12], and later work improved item tokenization and collaborative semantics through methods such as LETTER and LCRec [17, 22]. SIDReasoner builds on this line by treating SIDs not only as compact identifiers but also as objects an LLM can reason over.

Reasoning and reinforcement learning for recommendation. Reasoning in LLMs is often implemented through explicit intermediate steps, as

in chain-of-thought prompting [18]. Reinforcement learning can refine such behaviour when final outcomes are verifiable; SIDReasoner uses GRPO, which compares groups of sampled outputs without a separate value model [13]. In recommendation, recent work has explored explicit reasoning, such as Reason4Rec and ThinkRec, and latent reasoning, such as ReaRec [3, 21, 15]. Our study differs by focusing on the reproducibility and robustness of the SIDReasoner pipeline. We test whether the reported Stage 3 improvement can be recovered, whether explicit reasoning is useful at inference, and whether the approach transfers beyond the original Amazon domains.

Methodological extensions. Our extensions also relate to constrained decoding and parameter-efficient fine-tuning. In SID-based recommendation, catalogue validity is essential because a generated token sequence is useful only if it maps to an existing item. We therefore distinguish between enforcing valid SID paths during RL rollouts and only at inference. This distinction matters because constraints can improve output validity but may reduce exploration during policy optimization. We also compare full-parameter updates with LoRA, which adapts large language models through low-rank trainable parameters while keeping the base weights fixed [10]. These extensions are not proposed as new recommender architectures, but as probes of which implementation choices affect the reproducibility and practical behaviour of SIDReasoner.

3 Methodology

SIDReasoner formulates sequential recommendation as autoregressive generation over Semantic IDs (SIDs). Given a user interaction history, each item is converted into a fixed-length SID sequence, and the model generates the next item’s SID sequence. Figure 1 summarizes the pipeline.

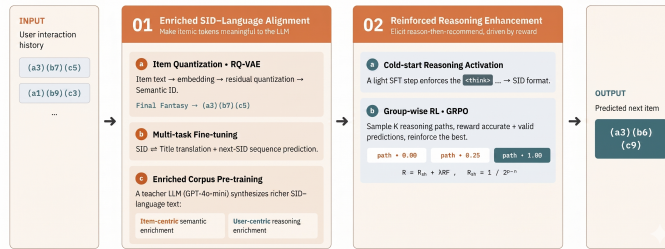


Fig.1: Overview of the SIDReasoner framework, consisting of enriched SID-language alignment and reinforced reasoning enhancement.

3.1 SIDReasoner Pipeline

The original paper presents SIDReasoner as a two-stage framework: enriched SID-language alignment followed by reinforced reasoning enhancement. The released training pipeline operationalizes this as three runnable phases. We therefore use Stage 1, Stage 2, and Stage 3 in the rest of the paper to refer to the

released checkpoints, which we describe below. Note that Stage 2 and Stage 3 together correspond to the paper’s reinforced reasoning enhancement stage, Stage 2 in Figure 1.

Each item is represented by three SID tokens, $SID(i) = \langle s_1, s_2, s_3 \rangle$, which are added to the Qwen3-1.7B tokenizer. Since these tokens are newly introduced, the model must learn how they relate to item metadata, natural language, and user interaction patterns.

Stage 1: Enriched SID-language alignment. Stage 1 aligns SID tokens with language through recommendation-oriented tasks, including next-item prediction and SID-title translation. The alignment data is further enriched with teacher-generated item-centric descriptions and user-centric reasoning narratives, exposing SID tokens to richer semantic and behavioural contexts.

Stage 2: Cold-start reasoning activation. Stage 2 is the first runnable phase of reinforced reasoning enhancement. Starting from the aligned checkpoint, the model is lightly fine-tuned to follow a reason-then-recommend format by generating a reasoning segment before predicting the target SID sequence. This produces the reasoning-activated checkpoint.

Stage 3: GRPO reinforcement learning. Stage 3 further optimizes the reasoning-activated checkpoint with GRPO [13]. For each user context, the model samples multiple reasoning-prediction trajectories and receives an outcome-based reward combining hierarchical SID prefix matching with catalogue validity.

3.2 Extension Mechanisms

Our extensions modify Stage 3 training and inference on Office Products. We vary the reward function, compare constrained and unconstrained RL rollouts, test LoRA-based updates [10], and evaluate whether explicit reasoning at inference improves ranking quality. Finally, we apply the same pipeline to a sampled Yelp Restaurants dataset [20] to evaluate transfer beyond the Amazon domains.

4 Experimental Setup

4.1 Datasets

We reproduce SIDReasoner on the three Amazon datasets used in the original paper: Video Games, Office Products, and Industrial & Scientific [5, 8]. Following the original setup, we use 5-core filtering, chronological user histories, maximum sequence length 10, and temporal train/validation/test splitting. Dataset statistics are reported in Appendix Table 2.

For the transfer extension, we use a sampled subset of the Yelp Open Dataset [20]. We restrict the data to Restaurants businesses and sample 10,000 users with at least five interactions. Yelp is substantially sparser and has a much larger item catalogue than the Amazon subsets, making it a stress test for domain transfer. Statistics are also reported in Appendix Table 2.

4.2 Metrics

We evaluate recommendation quality with Hit Ratio (HR@K) and Normalized Discounted Cumulative Gain (NDCG@K), for $K \in \{5, 10\}$. HR@K measures whether the ground-truth item appears in the top- K recommendations, while NDCG@K additionally rewards higher rank positions. Following the original paper, the Amazon reproduction uses full-item ranking rather than sampled negatives. For controlled Office Products comparisons, we report paired bootstrap intervals where relevant.

4.3 Baselines

The main reproduction compares our Stage 2 and Stage 3 checkpoints with the SIDReasoner values reported by He et al. [5]. Stage 2 evaluates the reasoning-activated model before reinforcement learning, while Stage 3 evaluates the GRPO update. We also reproduce SASRec [11] as a discriminative sequential recommendation baseline and compare it with the paper’s reported values. We do not rerun all baselines due to computational constraints. For Yelp, we compare SIDReasoner against SASRec trained on the same sampled data.

4.4 Implementation Details

All SIDReasoner runs use Qwen3-1.7B [19] and the released SID vocabularies. Stage 1 performs SID-language alignment, Stage 2 applies one epoch of reasoning activation, and Stage 3 uses a customized `ver1` GRPO setup [14]. Unless stated otherwise, Stage 3 uses batch size 256, rollout group size 16, KL coefficient 10^{-3} , learning rate 5×10^{-7} , and 10 configured epochs. Checkpoint identifiers refer to global optimization steps rather than epochs.

Experiments were run on Snellius with 4 H100 GPUs. Stage 3 was the main bottleneck because it combines rollout generation, multi-GPU training, GRPO optimization, and checkpoint merging, with some runs limited by a 48-hour wall-time. The Office Products extension starts from the same Stage 2 checkpoint and evaluates six RL variants: constrained-rollout control, no-validity reward, no-rollout-constraint reward, exact-only reward, LoRA no-validity, and LoRA no-rollout-constraint. Full-parameter variants use learning rate 5×10^{-7} ; LoRA variants use rank 32, alpha 64, and learning rate 5×10^{-6} . Full configuration details are reported in Appendix Table 3.

The `control` variant is an extension control rather than an exact paper-style Stage 3 reproduction because it uses constrained RL rollouts. The reproduced Stage 3 checkpoint remains the primary paper-style RL baseline. This distinction is important because the extension variants isolate reward and decoding choices rather than replace the reproduction checkpoint.

5 Results: Reproducibility

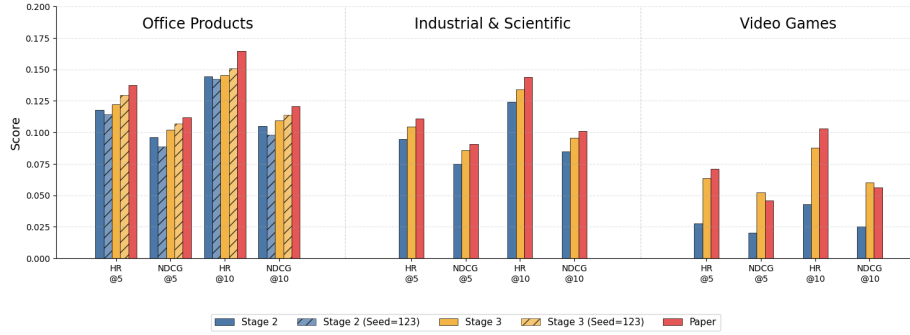


Fig. 2: SIDReasoner reproduction results on the three Amazon datasets. Reproduced Stage 2 and Stage 3 checkpoints are compared with the reported values across HR@5, NDCG@5, HR@10, and NDCG@10. Office Products additionally includes a second seed. Full results are reported in Appendix Table 4.

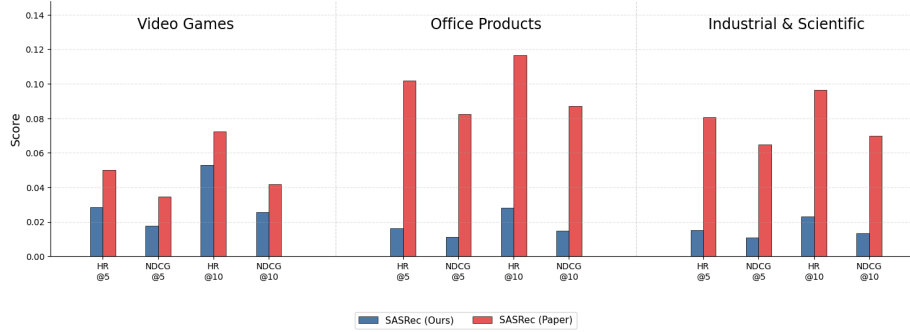


Fig. 3: SASRec reproduction results on the three Amazon datasets. The reproduced baseline is compared with the reported values across HR@5, NDCG@5, HR@10, and NDCG@10. Full results are in Appendix Table 5.

5.1 SIDReasoner

Figure 2 compares our reproduced Stage 2 and Stage 3 checkpoints with the results reported by He et al. [5]. The main trend reproduces well: Stage 3 improves over Stage 2 on all three Amazon datasets and metrics, with small gains on the primary Office Products seed but larger gains elsewhere. The strongest improvement is on Video Games, where HR@10 rises from 0.0428 to 0.0879 and NDCG@10 from 0.0251 to 0.0602.

The scores are only partially reproduced. On Office Products and Industrial & Scientific, our Stage 3 checkpoints trail the paper on all metrics. A second Office Products seed improves on the primary run (0.1508 HR@10, 0.1139 NDCG@10) but still falls short of the reported 0.1648/0.1208. The gap is smaller on Industrial & Scientific (0.1343/0.0956 vs. 0.1438/0.1010). Video Games is the exception:

our Stage 3 run trails the paper in HR but exceeds it in NDCG, meaning the target item is retrieved less often in the top 10 but ranked higher when it is.

Overall, the reproduction succeeds methodologically rather than numerically: RL consistently improves over the reasoning-activation checkpoint across all datasets, confirming that Stage 3 behaves as intended even where absolute scores fall short. Exact numerical agreement appears sensitive to initialization, training stochasticity, and implementation details not fully specified in the paper.

5.2 SASRec Baseline

Figure 3 reports the SASRec diagnostic baseline. Our results fall consistently below the paper, especially on Office Products and Industrial & Scientific, suggesting that part of the reproduction gap stems from preprocessing, training, or evaluation sensitivity rather than SIDReasoner itself. Since this discrepancy appears even in the supervised baseline, the remaining mismatch is unlikely to come from the RL stage alone. Instead, the full setup appears sensitive to preprocessing, split construction, baseline configuration, and evaluation details, making exact numerical reproduction difficult even when the qualitative trend holds.

Several practical issues likely contributed. The released scripts required dataset overrides, corrected reward-script paths, disabled W&B logging, environment fixes, and checkpoint merging; Stage 3 was also sensitive to checkpoint choice and wall time. Since most reproduction results are single runs, we do not apply significance testing or multiple-comparison correction; statistical uncertainty is instead addressed in the controlled Office Products extension, where paired comparisons are available.

Overall, the SASRec results indicate that preprocessing, initialization, optimization dynamics, or evaluation details can noticeably affect final metrics. Still, the reproduced baseline preserves the same ordering of dataset difficulty, making it a reliable reference for evaluating SIDReasoner’s improvements.

6 Extension

The reproduction shows that Stage 3 usually improves over Stage 2, but does not identify which training or inference choices matter most. We therefore extend SIDReasoner in three directions: Office Products reward and rollout ablations, inference-time reasoning, and transfer to Yelp Restaurants.

6.1 Reward Design and Rollout Constraints

All Office Products variants start from the same reproduced Stage 2 checkpoint. Table 1 summarizes the main validation findings. The constrained-rollout control regresses relative to Stage 2, decreasing NDCG@10 from 0.1789 to 0.1742. Removing the validity bonus partly recovers this loss, suggesting that the validity reward is less informative when rollouts are already constrained to valid

catalogue paths. The strongest validation result is obtained by removing rollout constraints during RL while restoring constrained decoding at inference, a pattern observed in both the full-parameter and LoRA variants.

The main conclusion is not that catalogue constraints should be removed entirely. Native unconstrained inference produces invalid or low-quality top-ranked SIDs. Instead, the most effective combination is unconstrained RL exploration with constrained decoding at evaluation.

Table 1: Summary of Office Products extension results. Validation rows use direct constrained SID ranking unless noted.

Model / comparison	Split	NDCG@10	Main takeaway
Stage 2 reference	Val.	0.1789	Reference
Control, constrained rollouts	Val.	0.1742	Regresses
No validity bonus	Val.	0.1772	Mostly recovers loss
No rollout constraints	Val.	0.1815	Better exploration
LoRA no rollout constraints	Val.	0.1817	Best validation result
Reproduced Stage 3, reasoning off	Test	0.1156	Direct SID ranking
Reproduced Stage 3, reasoning on	Test	0.1099	Reasoning hurts
LoRA no rollout constraints, reasoning off	Test	0.1204	Best frozen test result
LoRA no rollout constraints, reasoning on	Test	0.1157	Reasoning still hurts

6.2 Inference-Time Reasoning

We test whether generated reasoning improves ranking quality at inference. In the reasoning-on setting, the model generates a reasoning trace before constrained SID ranking. In the reasoning-off setting, the reasoning block is left empty and the model ranks SIDs directly. The result is consistently negative. On the Office Products test split, reasoning lowers NDCG@10 from 0.1156 to 0.1099 for the reproduced Stage 3 checkpoint and from 0.1204 to 0.1157 for the LoRA no-rollout-constraint variant. It also increases runtime substantially: direct inference takes roughly three minutes per model, while reasoning-on inference takes around 48–52 minutes. Confidence-gated reasoning reduces the number of generated tokens but does not clearly outperform direct SID prediction.

6.3 Transfer to Yelp Restaurants

Figure 4 reports the Yelp Restaurants extension. Since the authors did not provide a pre-trained RQVAE, we designed a category-frequency encoding that preserves the required three-level SID structure. A known limitation is that multiple items can share identical SIDs, corrupting the RL reward signal and likely explaining why Stage 3 does not improve over Stage 2. SASRec, operating directly on item IDs, is unaffected by this issue and outperforms both stages, highlighting that SIDReasoner’s effectiveness depends on high-quality semantic tokenization.

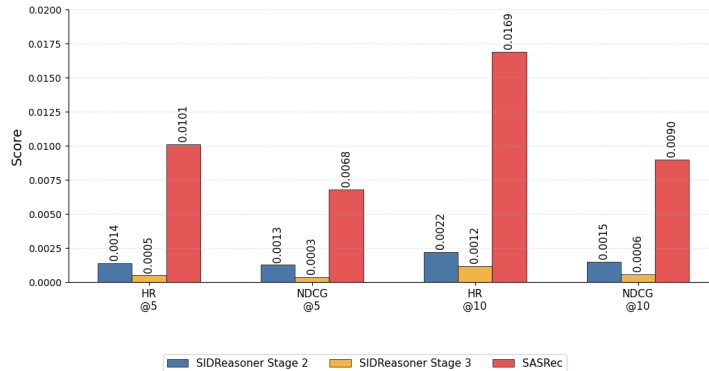


Fig. 4: Transfer results on the sampled Yelp Restaurants dataset. The figure compares SIDReasoner Stage 2, SIDReasoner Stage 3, and SASRec across HR@5, NDCG@5, HR@10, and NDCG@10. Full numerical results are reported in Appendix Table 6.

7 Discussion

RQ1: Reproducibility. The reproduction supports SIDReasoner’s main claim: GRPO-based Stage 3 training improves over the reasoning-activated Stage 2 checkpoint on all three Amazon datasets. However, exact reproduction is not achieved. The gap is largest on Office Products, smaller on Industrial & Scientific, and mixed on Video Games, where our model has lower HR but higher NDCG than the paper. The additional Office Products seed improves over the primary run but remains below the reported score, suggesting that stochasticity matters but does not fully explain the discrepancy.

RQ2: RL design. The Office Products ablations show that Stage 3 is sensitive to reward and decoding. A validity bonus is not clearly useful when rollouts are already constrained to valid catalogue paths. The strongest improvement comes from unconstrained RL exploration with constrained decoding at inference. This suggests that invalid or partially valid trajectories provide useful learning signals during policy optimization, while deployment should remain catalogue-constrained. The LoRA no-rollout-constraint variant is promising, although it is not a fully controlled comparison because learning rate and retained step differ from the full-parameter variants.

RQ3: Inference-time reasoning. Generated reasoning is not beneficial as an unconditional inference strategy. For both evaluated Office Products checkpoints, reasoning-on inference lowers ranking quality and substantially increases runtime. This does not prove reasoning is useless; a correct recommendation should not be interpreted as evidence that the generated reasoning trace was necessary or faithful. Future work should evaluate reasoning quality directly and design rewards that distinguish useful reasoning from redundant text generation.

RQ4: Domain transfer. The SIDReasoner pipeline transfers structurally to Yelp Restaurants, with all three stages running unmodified. However, results

quality seems tied to semantic tokenization, likely because category-frequency SIDs introduced vocabulary ambiguity. This does not undermine cross-domain transferability, but suggests that transfer is sensitive to SID quality, making robust tokenization a prerequisite for meaningful evaluation. Confirming this would require full RQVAE training on Yelp metadata, which compute constraints put outside this work’s scope.

What was easy. The released pipeline was easy to follow once the data and environment were prepared. The three operational stages map clearly onto the method described in the original paper: first align SIDs with language, then activate reason-then-recommend behaviour, and finally apply GRPO. The ranking metrics were also straightforward after checking the evaluation output order, since HR and NDCG are standard top- K recommendation metrics.

What was hard. The difficult part was obtaining a reliable end-to-end reproduction. The released code contained dataset-specific defaults and path assumptions that could silently lead to incorrect configurations, especially for non-default datasets. Stage 3 was particularly fragile because it depended on multi-GPU rollout generation, customized `verl` components, reward scripts, checkpoint merging, and wall-time limits. These issues did not invalidate the method, but made exact reproduction substantially harder than a standard supervised recommender.

Limitations. Our study is limited by compute and time. Most reproduction results are single runs, with only one additional Office Products seed. The LoRA and full-parameter variants are not perfectly controlled for learning rate or update count. The Yelp evaluation is limited by the use of category-frequency SIDs; a fair comparison would require RQVAE-generated SIDs. Finally, we evaluate generated reasoning mainly through ranking performance and cost, not through human or automatic faithfulness evaluation.

8 Conclusion

We reproduced and extended SIDReasoner, a generative recommendation framework that reasons over Semantic IDs. On the three Amazon datasets, we recover the main qualitative trend that Stage 3 GRPO training improves over the Stage 2 reasoning-activated checkpoint. However, we do not fully match the reported scores, and the Office Products seed experiment shows that stochasticity explains only part of the gap. Our extensions show that Stage 3 performance depends strongly on reward and decoding choices: unconstrained RL exploration combined with constrained inference is more promising than constraining rollouts throughout training. In contrast, explicit generated reasoning at inference reduces ranking quality and greatly increases runtime. The Yelp Restaurants experiment further shows that SIDReasoner does not automatically transfer to a different recommendation domain. Overall, SID grounding and outcome-based RL are useful, but their gains depend on implementation choices, decoding constraints, and dataset characteristics.

A Contributions

- **Alexandra:** Extended SIDReasoner on Yelp dataset, ran SASRec baseline, and worked on the poster and the report.
- **Thomas:** Designed and ran the Office Products extension experiments, including reward, rollout, LoRA, reasoning ablations and worked on the report.
- **Erin:** Implemented and ran the SIDReasoner reproduction pipeline for the original seed for Datasets Office products and Industrial & Scientific. Worked on the poster and the report.
- **Mattia:** Carried out the reproduction of the Office Dataset with extra seed, ran the reproduction of the video games dataset with original seed and worked on the report.

B Additional Result Tables

This appendix reports the Amazon dataset statistics, extension configurations, and full numerical results corresponding to the figures in the main text. The main paper visualizes the most relevant trends, while the tables below preserve the exact HR@5, NDCG@5, HR@10, and NDCG@10 values.

Table 2: Dataset statistics for Amazon and Yelp.

Dataset	Items	Interactions	Train	Validation	Test
Video Games	3,858	61,417	49,133	6,142	6,142
Office Products	3,459	48,656	38,924	4,866	4,866
Industrial & Scientific	3,686	45,325	36,259	4,533	4,533
Yelp Restaurants	46,465	67,917	47,917	10,000	10,000

Table 3: Office Products Stage 3 extension configurations.

Variant	Reward	RL rollout	Update	Steps
control	Hierarchical + validity	Constrained	Full	500
no_validity	Hierarchical	Constrained	Full	500
no_constrained	Hierarchical	Unconstrained	Full	500
exact_only	Exact item only	Constrained	Full	500
lora_no_validity	Hierarchical	Constrained	LoRA	200
lora_no_constrained	Hierarchical	Unconstrained	LoRA	200

Table 4: Full reproduction results of SIDReasoner using Qwen3-1.7B. HR denotes Hit Ratio.

Dataset	Model	HR@5	NDCG@5	HR@10	NDCG@10
Office Products	Stage 2	0.1180	0.0963	0.1443	0.1048
	Stage 2 (Seed = 123)	0.1145	0.0887	0.1426	0.0979
	Stage 3	0.1223	0.1019	0.1453	0.1093
	Stage 3 (Seed = 123)	0.1295	0.1070	0.1508	0.1139
	Paper	0.1373	0.1119	0.1648	0.1208
Industrial & Scientific	Stage 2	0.0946	0.0750	0.1244	0.0846
	Stage 3	0.1046	0.0860	0.1343	0.0956
	Paper	0.1109	0.0905	0.1438	0.1010
Video Games	Stage 2	0.0275	0.0202	0.0428	0.0251
	Stage 3	0.0635	0.0524	0.0879	0.0602
	Paper	0.0710	0.0460	0.1031	0.0563

Table 5: Full SASRec reproduction results compared with the values reported in the original paper.

Dataset	Model	HR@5	NDCG@5	HR@10	NDCG@10
Video Games	SASRec (Ours)	0.0284	0.0176	0.0530	0.0256
	SASRec (Paper)	0.0501	0.0345	0.0723	0.0416
Office Products	SASRec (Ours)	0.0164	0.0113	0.0282	0.0149
	SASRec (Paper)	0.1019	0.0824	0.1167	0.0871
Industrial & Scientific	SASRec (Ours)	0.0153	0.0109	0.0229	0.0132
	SASRec (Paper)	0.0807	0.0647	0.0964	0.0697

Table 6: Full results on the sampled Yelp Restaurants dataset.

Model	HR@5	NDCG@5	HR@10	NDCG@10
SIDReasoner Stage 2	0.0014	0.00126	0.0022	0.00151
SIDReasoner Stage 3	0.0005	0.00035	0.0012	0.00057
SASRec	0.0101	0.0068	0.0169	0.0090

C Additional Office Products Ablations

Table 7 reports the full Office Products ablation results. The main text reports a compact summary, while this appendix table includes the full validation and test metrics for the reward, rollout, LoRA, and inference-time reasoning comparisons.

Table 7: Full Office Products ranking results. “On” generates a reasoning trace before constrained SID ranking; “Off” uses an empty reasoning block and ranks SIDs directly.

Dataset	Split	Ablation / model	Reasoning	HR@5	NDCG@5	HR@10	NDCG@10
<i>Exploratory reward/rollout ablations; direct SID ranking</i>							
Office Products	Validation	Stage 2 reference (constrained)	Off	0.1969	0.1697	0.2254	0.1789
	Validation	Control (constrained rollouts)	Off	0.1954	0.1663	0.2197	0.1742
	Validation	No validity bonus	Off	0.1987	0.1707	0.2189	0.1772
	Validation	No rollout constraints (constrained eval.)	Off	0.2008	0.1760	0.2180	0.1815
	Validation	Exact-only reward	Off	0.2002	0.1736	0.2164	0.1789
	Validation	LoRA: no validity bonus	Off	0.1985	0.1706	0.2219	0.1782
	Validation	LoRA: no rollout constraints (constrained eval.)	Off	0.2008	0.1750	0.2215	0.1817
	Validation	No rollout constraints (native unconstrained eval.)	Off	0.1958	0.1714	0.2104	0.1763
	Validation	LoRA: no rollout constraints (native unconstrained eval.)	Off	0.1944	0.1691	0.2119	0.1749
<i>Fixed inference-mode comparison; full validation</i>							
Office Products	Validation	Reproduced Stage 3	Off	0.1944	0.1715	0.2092	0.1763
	Validation	Reproduced Stage 3	On	0.1909	0.1679	0.2028	0.1718
	Validation	LoRA: no rollout constraints	Off	0.1993	0.1750	0.2176	0.1809
	Validation	LoRA: no rollout constraints	On	0.1975	0.1727	0.2139	0.1779
<i>Frozen comparison; official test split</i>							
Office Products	Test	Reproduced Stage 3	Off	0.1289	0.1107	0.1441	0.1156
	Test	Reproduced Stage 3	On	0.1245	0.1067	0.1342	0.1099
	Test	LoRA: no rollout constraints	Off	0.1342	0.1144	0.1525	0.1204
	Test	LoRA: no rollout constraints	On	0.1301	0.1105	0.1463	0.1157

Notes. Each row evaluates 4,866 examples. All non-native rows use constrained beam-10 SID decoding. The two rows labelled “native unconstrained” use unconstrained beam-10 decoding and are included because they are part of the rollout ablation. The first validation block and the fixed inference-mode validation block use different evaluators, so rows with the same model name should only be compared within their labelled block.

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